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A Mojo-Native Differentiable, GPU-Native Astrodynamics Substrate

Layered Implementation Roadmap from Compiler Primitives to Operational Propagator

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> **Reconstruction notice.** This page is a summary rebuilt from fragments of the original conversation. The linked PDF is a PDF render of *this reconstruction*, not the original monograph. To replace it with the full source, drop the real `.pdf` into `/pdfs/` using the same filename.

§ Abstract

This monograph specifies a complete implementation roadmap for a high-fidelity differentiable orbit propagator built on the Mojo programming language and the broader Modular Platform, structured as a layered architecture from compiler-level primitives upward to a production-grade propagator. The structural difference from a JAX-equivalent roadmap is significant: where JAX provides production-grade automatic differentiation, ODE integration, and ecosystem libraries on day one, Mojo as of 2026 requires that several of these layers be built rather than imported. This shifts the project shape from "compose existing components" to "build the foundations, then compose them," with both a higher engineering burden and a higher strategic upside in the form of greater architectural control and a publishable contribution to the Mojo ecosystem.

§ Structural difference from JAX

The intended audience is the technical lead who would own this work and is willing to absorb the additional foundational effort in exchange for IR-level optimization control, AOT deployment, and the language properties that distinguish Mojo from Python-based alternatives.

The structure mirrors the parallel JAX-stack roadmap (No. 03); identical sections in the two documents address the same engineering problem. The comparative read is the point — every layer can be evaluated against its JAX analog and decided on the explicit trade.

§ What must be built that JAX gives away free

- **Forward-mode AD over numerical primitives.** Trait-driven via `HasPrimal` /dual-number lifting; spec'd in detail in companion PRD v12.
- **Reverse-mode AD for whole-trajectory adjoints.** Graph IR with explicit checkpointing; the heaviest engineering item below the propagator.
- **An adaptive ODE integrator family.** RK 8(7), Gauss-Jackson, and a symplectic option; built against the GLMSpec abstraction so coefficients are derived at compile time, not hand-coded.
- **SIMD/GPU kernels for the dynamics RHS.** Mojo's strengths show here; this is what makes the foundational work worth it.
- **BLAS/LAPACK shim or replacement.** Decision point — wrap existing C libraries via FFI, or write tile-based kernels in Mojo. The roadmap argues for hybrid: wrap for linear algebra in the optimizer outer loop, native kernels for the dynamics inner loop.

§ Strategic case

The compounding upside: each foundational layer built is a publishable artifact in a young ecosystem. A Mojo-native SOFA reimplementation, a Mojo orbit propagator with SIMD/GPU benchmarks, a Mojo AD-enabled GLMSpec integrator family — each lands in a domain where no comparable reference exists. The JAX roadmap delivers operational capability; the Mojo roadmap delivers operational capability *plus* three to five papers and a position in the ecosystem.

❦

§ Status & provenance

Third of three monographs from conversation 2308b2c5 on 2026-05-08. Read in sequence with Nos. 02 (the technical case) and 03 (the JAX-stack roadmap). The PRD v12 (No. 09) provides the architectural foundation this roadmap assumes.